#25

Group Number

MyEyes
AI Assistive Glasses for the Visually Impaired

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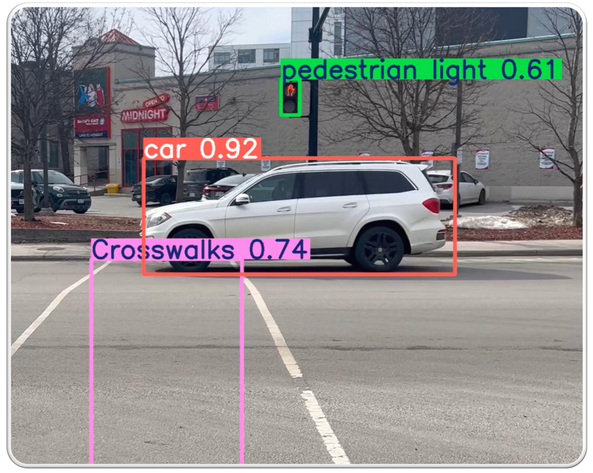
Problem Statement

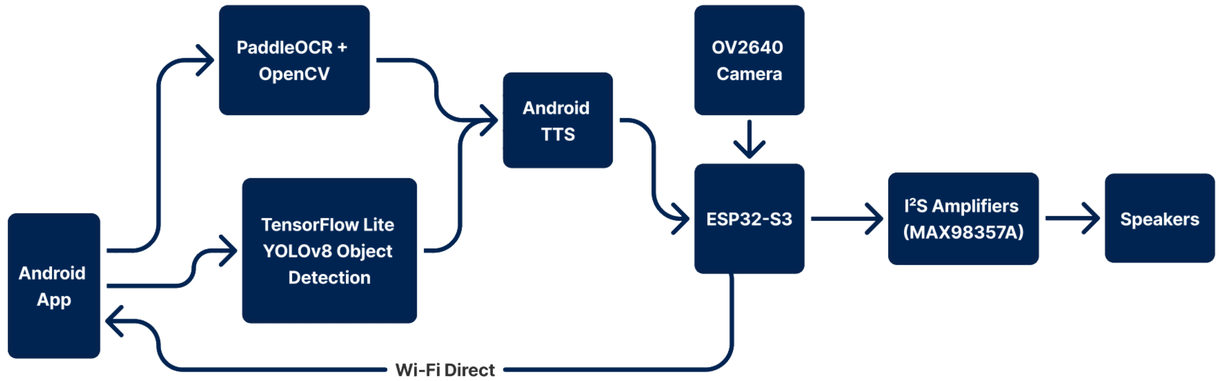
Over 2.2 billion people worldwide have some form of visual impairment. Traditional assistive devices often lack real-time environmental awareness and are cumbersome to use in daily activities.



Our Solution

Smart glasses that combine computer vision with audio feedback to provide real-time environmental information to users.

System Design Diagram



Technology Stack

ESP32-S3 | PlatformIO | Python | Android Studio | Pytorch | Roboflow

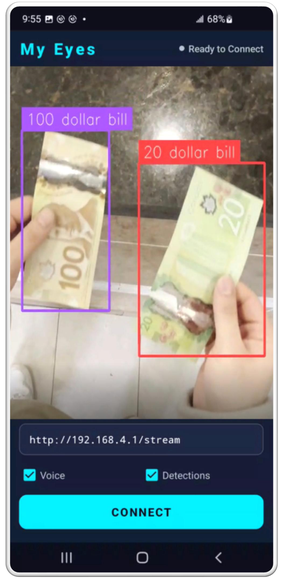
Performance Metrics

Detection Type	Confidence	Latency
Vehicle	94%	<85ms
Readable Text (OCR)	93%	<100ms
Crosswalk	81%	<93ms
Currency	91%	<95ms
Pedestrain Light	86%	<83ms

Android Companion App

The MyEyes mobile app acts as the central processing hub. It offloads heavy AI computations from the glasses to preserve battery life and enable more sophisticated analysis of environmental information.

- Wireless Communication
- AI Inference Engine
- Command Transmission



Future Enhancements

- Facial Recognition, to identify familiar faces
- ML improvements for better accuracy
- Multi-language support for global accessibility
- GPS integration for turn-by-turn audio directions
- Optimize 3D design and component size
- Add microphone for real time assistance
- IOS Compatibility for broader market appeal

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